Waterloo Potter's Workshop Electric Kiln Manual

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1. Introduction to the electric kilns

Waterloo Potter's Workshop has 4 electric kilns. Three of these have computerized controllers while the fourth is a manual kiln with a kiln sitter.

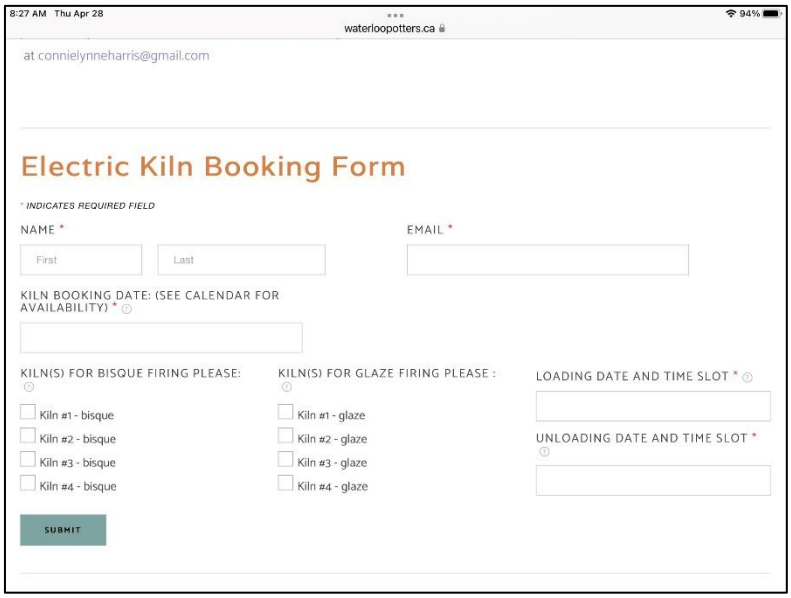
The Electric Kiln Convener is responsible for ensuring that members have timely access to booking electric kilns, kiln furniture is maintained and replaced as needed, kiln room is kept clean and safe and kiln maintenance is performed at appropriate times so that kiln operation is optimal.

A WPW member is assigned as responsible for Electric kiln Maintenance and repairs to ensure electric kiln safety, performance and (high) availability to workshop members, and reports to the Electric Kiln Convener. The Pottery Supply House (PSH) technicians are called in on an as-needed basis to provide inspection and repairs that cannot be done by the WPW kiln maintenance person. The position descriptions provide more details.

2. On-line booking and payment of electric kilns

Booking

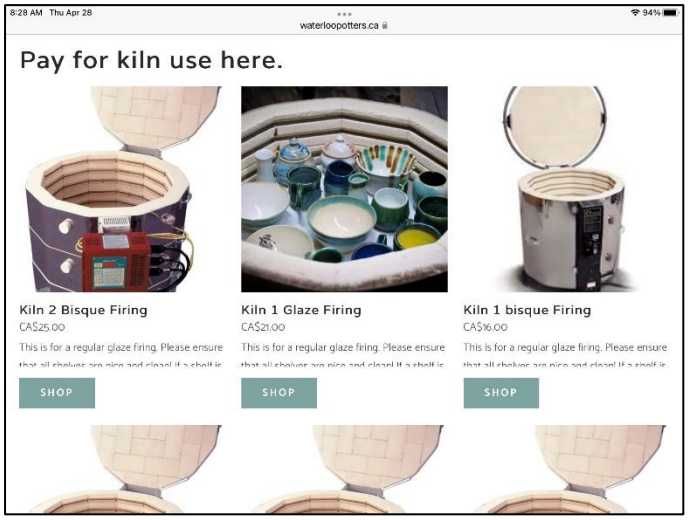
In 2020, an online form was implemented for booking kilns. You must log into the members only section of the website and scroll down to "Electric Kilns". The facilities calendar appears first, so that you can check for available dates for each kiln. Scroll down to the Electric Kiln Booking form, shown here. You must request the kiln #, booking dates including loading and unloading times, and your email. The Electric kiln convener will confirm your bookings within 24 hours by sending calendar invites to your email or will contact you if there is a conflict or you have filled the form incorrectly. Generally, bookings are accepted on a first-come, first-serve basis, but we also need to ensure that kilns are not being monopolized by a few people.



The Electric Kiln Convener tracks utilization patterns. It is advisable to book your desired kiln dates 6-8 weeks ahead of the spring and fall sales. You should not expect to load before 6 p.m. the evening BEFORE your kiln firing day and must be unloaded by 6 p.m. the day following. Unless you plan to share kilns with a colleague and will be loading/unloading together, avoid booking these activities at the same time as another user, because of the limited workspace.

3. Payment for Kilns

The Electric Kiln Convener documents the kilns booked and the user's name and submits this to the WPW Bookkeeper every 2 months. The Bookkeeper reconciles the bookings with the payments and will contact users who have not submitted their payments. To pay for your kilns, scroll down on the Electric Kiln booking page to find the "Pay for your Kilns here" section as shown here, select kiln and type of firing by clicking on "shop", enter the firing date and your personal information and proceed to payment. You will receive a receipt by email.



4. Kiln Room Safety

Ventilating the Kilns

Each kiln has a ventilation fan that exits at the side near the bottom to remove hazardous fumes to the outside of the building. They are wired so that kilns 1 & 2 and 3 & 4 kilns are exhausted together, whether both kilns are being used or not. You must turn them ON when firing and turn OFF when firing is complete. Leaving the kiln fans running can cause faster cool down of the kilns, which may interfere with the results of some glazes.

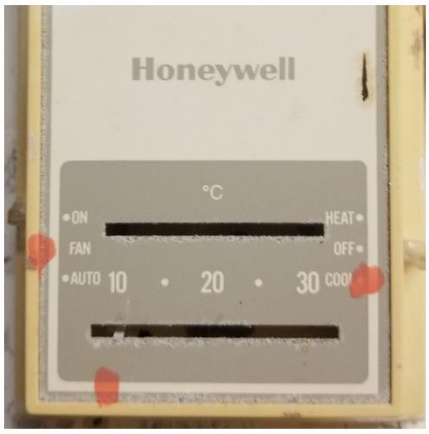
Ventilating the kiln room

You must turn on the intake and output fans using central thermostat on the wall beside the electrical panel box, between kilns 2 and 3. These two fans must be running when the kilns are firing to reduce heat and hazardous fumes from spreading to the rest of the workshop. They are both wired and synchronized so that they can be controlled by the thermostat-type switch but the individual switch at each fan must be left turned on. There is a reminder sign at each of the fans. Instructions by the thermostat (shown here) outline how to turn these fans on and off centrally, and to remind kiln users that they need to check both of them. In general you should turn room fans ON when firing and turn OFF when firing is complete, although during hot summer days they could be left on to help cool the basement.

This thermostat is synchronized with 2 room fans to ventilate the kiln room during kiln firings. It does not affect the heating system.

It is "ON" when in the positions shown by the pink dots. Please make sure that the intake fan above the yellow kiln shelves and the large fan by the outside basement door are also running. Their switches should always be in the "ON" position, but they won't run unless this thermostat is also turned "ON" to "Cool".

Please do NOT turn them "OFF" using their switches. Instead using the thermostat, flip the right hand switch up to "off" and leave the temperature setting as is. Both fans should turn off.



Preventing Damage to Shelves:



Did you know? 6 Stackable Tile/Plate Setters are stored under the "Please fire me" shelf.
You can use them to maximise space in your kiln, but only on the bottom shelf in the kiln after placing extra cookies under the area where they will sit (but not touching elements). You need to support the additional weight to prevent breakage of the shelf.



NOTE: if you break a kiln shelf or post due to dropping it or not supporting properly as in the example above, you must order and pay for the replacement. If it breaks during firing due to unknown reasons, the workshop will pay for the replacement.

Use of Sink in kiln room

This sink does NOT have a clay trap under it.

- It is OK to use these taps to get water into a bowl to wipe pots or for clean-up, but you must take your dirty water back upstairs and dump into the mud room sink.
- **DO NOT** leave the plug or anything in the sink that would block the drain, as overflow water comes into the sink from the heating system and must drain away
- **PLEASE DO NOT** put kiln shelves on this sink to chip glaze or kiln wash – use the cart in the corner under the stairs
- **Never** let any water with clay residue (from wiping greenware or shelves) go down the drain
- Washing glaze drips, cracked kiln wash, or clay water down the drain could block the drain, which would be a disaster, requiring $$$$$$$$ and digging up the basement floor

Water on the floor in the kiln room

We CANNOT fire the kilns if the basement floor is wet.
If it has rained recently, please check the red pail under the pipes to the right of the sink when you are in the kiln room.
If there is any water in it, please empty it into the sink and replace the pail to the position under the pipe, and email the kiln convener.
If we keep emptying it when there is water dripping, hopefully we can prevent a wet basement floor and a dangerous situation for those wanting to fire kilns.
If you must mop up the water, be sure to only stand on dry concrete and never stand in the water. NEVER TOUCH THE KILNS or CONTROLLER WHILE STANDING IN WATER.

5. Using the Computer-controlled kilns (Kilns 2, 3 and 4)

The following WPW Firing Procedure is laminated and hangs from a hook at the right-hand side of the kiln room countertop. Please consult the manufacturer's Manual for the Bartlett V6-CF Series 700 Controller found in the kiln booklet binder. It is also available at this location: https://archive.org/details/manuallib-id-2716514

Loading

- ☐ Check the kiln bottom for any dust/sand/debris left by previous user. Vacuum again if necessary (see UNLOADING below for instructions).
- ☐ When loading, be careful not to bump thermocouples and keep all pots at least 1 cm from walls.
- ☐ For a bisque firing, you can flip the shelves and use the bare side. This helps straighten the slumping of the shelves, especially the kiln 2 shelves. For glaze firing, use of cookies is optional but recommended to prevent glaze from adhering to shelves.
  Always examine and dust the shelves top and bottom, removing any kiln wash that is adhered to the underside of shelves.
  Clean the top and bottom of the posts before inserting. Cement Blocks beside each kiln are ideal for scraping both ends of posts before inserting into kilns. Kiln wash that is left on these posts will adhere to the shelves, causing the posts to stick and potentially drop onto your pots when removing the shelves. When re-fired, if not removed, this residue can drop into your pots and ruin the piece.
- ☐ **Don't forget to load the appropriate Witness Cone Packs at the peep holes.** Please make new cone packs to replace the ones you have used.

Firing

- ☐ Check that the basement fire exit door opens, but keep it shut during the firing. Check for flammables near kiln and clear away.
- ☐ Close kiln lid using safety handle (or kiln will not stay on). Close peep holes.
- ☐ In panel box, check that numbered switch for the kiln being used is turned on. Turn on the room fan and kiln fan.
- ☐ Turn on power switch on side of the control box. For kiln 2 this is on the wall, for kilns 3 and 4 it is attached to the kiln. The digital display will read WAIT or it may show length of time of last firing and current temp. If so, press ENTER to get the IDLE message, ready to accept your program. The temperature will flash with Idle.



There are two methods of firing:

- **Cone Fire** is the quick, easy method of programming for ceramics and is described below.
- **Vary Fire** (also known as ramp-hold) is the versatile programming method for writing your own profiles for glass, jewelry and ceramics, for advanced kiln-users.

Cone Fire procedure: While temperature is flashing:

- ☐ Choose one of the firing speeds
  - **Slow Bisque** (RECOMMENDED esp. for large or thick items) e.g. Cone **06** ~12 hrs
  - Fast Bisque e.g. Cone **06** ~9 hrs
  - **Slow Glaze** (RECOMMENDED) e.g. Cone **6** ~9 hrs (NEVER fire above cone 7)
  - Fast Glaze (not recommended as the clay may not vitrify and it may damage the kiln) e.g. Cone **6** ~5 hrs
  - ☐ **Press ENTER**. Display will confirm your choice, then ask for the cone #.
- ☐ Type in cone # (to correct, press 0 three times, press ENTER) (e.g. Cone 06 for bisque and 6 for glaze, but this varies based on the member's preference and glazes)
- ☐ **Press ENTER**
- ☐ **HOLD/ SOAK TIME:** This extends the time at **<u>the end of firing</u>** without increasing the actual kiln temperature. However, due to heat work, it may push the cone achieved to a higher cone, e.g. a Cone 5 glaze firing with a 15 minute hold may achieve a Soft cone 6. You must look back to see how the kiln is firing to determine how much to hold; usually between 10 and 20 minutes are used by WPW members. Set soak/hold time or leave at 0.
- ☐ **Press ENTER.** IDLE TC 2 will display briefly; then current temperature will be flashing.
  Note: The DELAY program is NOT allowed at WPW as you need to be in the building for the first hour of firing.
- ☐ ***Pre-heat option (like a warming oven):*** *With any of the CONE FIRE modes, a preheat stage is available* ***<u>before firing,</u>*** where *the temperature ramps up at 60°F/hour to 200°F* (93.3°*C) and then holds at 200°* (93.3°*C) for the amount of time programmed then goes automatically into the cone fire program. This can be used for drying ware.*

If you start at a room temperature of 70°F (21°C), then it will take just over 2 hours to reach 200°F (93.3°*C) at which time the hold time will start, although we generally find that it adds 1 hour to the actual preheat time. So if you program a 4 hour preheat, it actually adds about 5 hours to the firing time.*

Preheat is automatically set to zero during cone fire programming and at the end of each firing. If a preheat stage is desired, it must be reprogrammed for each firing.

Preheat Example: To set a preheat time of 2 hours. (Remember: the temperature must be flashing to start the programming.)

- *Step 1: Press "Other"* until *PrHt shows. If PrHt does not show on the display, even after cycling through the options, it means that CONE FIRE mode has not been selected. Exit the menu and select a CONE FIRE speed, then return to the Other menu.*
- *Step 2:* Press *ENTER* so that *Preheat has been selected; enter the time you want to hold the temperature at 200°F* (93.3°C) *(in this ex. 2 hours)*
- *Step 3: Press 2,0,0 so that 2.00 displays. Numbers to left of decimal point are hours, to the right of decimal point are minutes. If you type a wrong number, press zero 4 times, then type the correct number.*
- *Step 4 Press ENTER CPL flashes then the current temperature CPL flashes several times indicating the 2 hour preheat time has been accepted. The current temperature then flashes in the display.*

☐ **"Review program"** can be used before or during firing to check on what you selected.

☐ **Press START/STOP** to begin firing. Display will flash ON and show current temperature. If this does not happen press START again. The display will tell you the target temperature e.g. cone 06 is 998°C; for cone 6 is 1223°C.

Regardless of the program, you must be present for the first hour and the last 2 hrs of the firing.

☐ Check the temperature every hour that you are present and record in your kiln firing form. Display flashes with current temperature and type of firing.

☐ As temperature approaches the goal, check the cone packs frequently. Use dark safety glasses to protect your eyes! Open the peeps with a metal tool, and blow into the peep hole to visualize the cone pack.

Turning off

☐ At end of firing, the computer display should read CPLT, while flashing with temperature and length of time of firing. Complete firing record sheet in kiln binder. If you had to use a manual stop rather than allowing the computer to stop, please record this.

☐ Turn off kiln power switch on side of front box on kiln. **Turn off kiln fans** to prevent rapid cooling; they can contribute to up to 1/3 faster cooling than allowing the kiln to cool naturally. This can negatively impact your glazes and pots, causing pinging and crazing, possibly even cracking. Turn off room fans unless it is a very hot summer day and leaving them on will keep the building cooler.

Cooling Procedures for Computer Controlled Kilns

Our kilns take approximately 20-24 hours to cool to the point that it can be safely emptied, but this is variable depending on the size of the load, type of firing, number of kilns in use and the time of year.

Unloading

Usually the next day:

☐ Turn on kiln to read temp., crack lid and open the peeps when the temperature is about 100° C or 200° F. Kiln manufacturers recommend that you not open the lid wide until the temperature is down to 52° C (125° F) or even room temperature. Opening the kiln when it is too hot can cause damage to the kiln and the pots, especially glazed pots which may craze or lose their vibrance if the kiln is opened when too hot. Use gloves to protect your hands. Be careful not to bump thermocouples when unloading.

☐ You must unload and clean up the kiln before 6 pm so the next load can begin then. In the kiln firing log, complete the diagram of the witness cone pack results, hours of firing and the status of the shelves plus any maintenance you have done on them.

☐ If there is glaze or crackled kiln wash on a kiln shelf, please follow kiln furniture maintenance instructions.

☐ Record which shelves you used, and what maintenance was performed.

- To use the central vacuum, first unravel the hose so that it is full length with no sharp kinks. Use two hands to insert the hose into/ remove from the outlet. Remove bottom kiln shelves and use the brush attachment to gently vacuum out the elements and bricks from top to bottom, and the kiln floor. Feel around the edge of the kiln floor to be sure that no debris has been left behind. Replace the kiln stilts in the correct position and replace the bottom shelves. Close the lid and peeps. Check off that you have vacuumed the kiln.
- IF there are problems with kiln firing or you have bumped a thermocouple, please notify kiln maintenance person and cc: the kiln convener.
- Notify the kiln convener regarding shelf warping or anything else that needs seen to.

Kiln Convener and Kiln Maintenance person contact information is posted on the bulletin board at the bottom of the stairs.

6. Vari-fire Programs with the Bartlett Controllers

The V6-CF has 6 vary-fire user programs to store and reuse. Each program has from 1 to 8 segments (2 – 16 segments if 16-S option is on). Each segment has a firing rate, a soak temperature, and a hold time. It is best to write out the firing profile that you plan to program before you begin programming. To ramp at the maximum rate either up or down, enter a rate of 9999. To program a down ramp, you enter the rate/hour then a temperature below the previous segment temperature. NOTE: The first ramp rate of any user program must be an increasing ramp which means the segment temperature must be greater than the starting kiln temperature. Never overwrite some one elses program on the WPW computers. For example one program is set to fire the kilns after new elements are installed, another program is used for slow cool glaze firing. If you are interested in using a vari-fire program, please speak with June Macdonald.

7. Using the Manual R-85 Kiln with LT-3K kiln sitter

The manual "kiln sitter" kiln is for use by members to fire, but also to provide a learning experience for those members considering purchase of a used or similar kiln for their home use. Small Pyrometric Cones or Bars are used in kilns with a Kiln-Sitter®. This mechanical device is designed to cut off electricity to the heating elements in the kiln when a cones bending temperature has been reached. As heat is applied to the kiln, the cone/bar softens and then bends under the weight of a metal sensing rod. In a properly adjusted KilnSitter®, when the cone/bar has bent to a 90-degree angle, the sensing rod will have moved down enough to release a weight outside the kiln. This action cuts off electrical power to the heating elements. (Orton Cones and Firing Practical Guide 2019).

The following WPW Firing Procedure is laminated and hangs from a hook at the right-hand side of the kiln room countertop. Please consult the manufacturer's Manual for the Kiln-Sitter in the kiln booklet binder.

Loading

- ☐ Check the kiln bottom for any dust/sand/debris left by previous user. Vacuum again if necessary (see UNLOADING below for instructions).
- ☐ When loading, be careful not to bump thermocouples and keep all pots at least 1 cm from walls.
- ☐ Always examine and dust the shelves top and bottom, removing any kiln wash that is adhered to the underside of shelves.
- ☐ Clean the top and bottom of the posts before inserting. Cement Blocks beside each kiln are ideal for scraping both ends of posts before inserting into kilns. Kiln wash that is left on these posts will adhere to the shelves, causing the posts to stick and potentially drop onto your pots when removing the shelves. When re-fired, if not removed, this residue can drop into your pots and ruin the piece.
- ☐ Put the correct kiln sitter cone (found on top shelf) inside the kiln sitter in the prongs:
    - On the outside, lift the grey latch button and hook the prong over it (shown under the thumb in photo). This opens the kiln sitter's claw inside.
    - Put the kiln sitter cone number side down centred across the two kiln sitter cone supports, with the auto turn off (single prong) above. This should hold the latch up.





- ☐ For glaze firing, use of cookies is optional but recommended to prevent glaze from adhering to shelves.

- ☐ Don't forget to load the appropriate Witness Cone Packs at the peep holes. Please make new cone packs to replace the ones you have used.

Firing



NEW January 2023: You must stay at the workshop until the kiln is set to HI (high). You can then leave but must return for the last 2 hours of firing. If you choose to leave for an hour or so, you will have to estimate how long the total firing will take based on previous firings documented in the log book.

- ☐ In panel box, check that the numbered switch for the kiln is on.
- ☐ Turn on room fan and kiln fan.
- ☐ Check for flammables near kilns. Check fire escape door. Close peep holes on kiln and latch the lid.
- ☐ Make sure the timer on the sitter is 30 mins to 1 hour higher than the time anticipated to fire (e.g. set to 12 hours in this photo)



- ☐ Push in the time sitter switch to turn the power on to the kiln. The red light will come on.
- ☐ The OFF setting is shown in the correct position (a little to the left of 12 o'clock) to engage each setting. There is a pink dot of nail polish on the kiln to show the correct placement. Switch the elements on to LO turning counter-clockwise. You should feel a click as the position is engaged, and the kiln motor will rev a little.
- ☐ The LO setting can be used to preheat and remove moisture from greenware for a selected number of hours if desired. Keep peepholes open for that purpose to allow the water vapour to escape.



- ☐ In general, each hour, turn the elements up by 2, so LO, 2, 4, 6, HI. (You can see the pattern of the previous firings in the kiln binder.)
- ☐ Each hour, check and record the temp from the pyrometer mounted on the wall. Note: this pyrometer is not too accurate, but it does tell you if the kiln is working!
- ☐ Record firing in kiln binder and keep notations throughout the firing.

Turning off

- ☐ Leave on HI until the kiln sitter switches off the kiln, then turn the element knob back to OFF position as shown.
- ☐ Turn off room fan and kiln fan.

Unloading

Usually the next day:

- ☐ Turn on kiln to read temp., crack lid and open the peeps when the temperature is about 100° C or 200° F. Kiln manufacturers recommend that you not open the lid wide until the temperature is down to 52° C (125° F) or even room temperature. Opening the kiln when it is too hot can cause damage to the kiln and the pots, especially glazed pots which may craze or lose their vibrance if the kiln is opened when too hot. Use gloves to protect your hands. Be careful not to bump thermocouples when unloading.
- ☐ You must unload and clean up the kiln before 6 pm so the next load can begin then. In the kiln firing log, complete the diagram of the witness cone pack results, hours of firing and the status of the shelves plus any maintenance you have done on them.
- ☐ If there is glaze or crackled kiln wash on a kiln shelf, please follow kiln furniture maintenance instructions.

- Record which shelves you used, and what maintenance was performed.
- To use the central vacuum, first unravel the hose so that it is full length with no sharp kinks. Use two hands to insert the hose into/ remove from the outlet. Remove bottom kiln shelves and use the brush attachment to gently vacuum out the elements and bricks from top to bottom, and the kiln floor. Feel around the edge of the kiln floor to be sure that no debris has been left behind. Replace the kiln stilts in the correct position and replace the bottom shelves. Close the lid and peeps. Check off that you have vacuumed the kiln in the log.
- IF there are problems with kiln firing or you have bumped a thermocouple, please notify kiln maintenance person and cc: the kiln convener.
- Notify the kiln convener regarding shelf warping or anything else that needs seen to.

Kiln Convener and Kiln Maintenance person contact information is posted on the bulletin board at the bottom of the stairs.

8. Kiln forms and recording your firing in the kiln binders

A peer evaluation system was implemented in 2020. Kiln shelves are numbered and recorded by the user in a Kiln Firing Log, and then evaluated by the next user for being in a glaze free condition. The goal is to maintain the shelves in good condition and to document the firings in a standardized format. There are separate forms for the manual kiln (#1) and for the computer kilns (#2,3,4), images shown below.

COMPUTERIZED KILN LOG

WPW Electric Kiln Firing Log

Date_____ Kiln # ____ Potters Firing_____

Load size: Heavy___ Medium___ Do not fire a partially filled kiln as it is hard on the elements and wastes electricity.

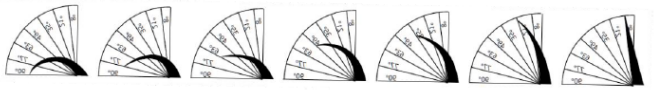
Program: Fast Bisque ___ Slow Bisque___ Fast Glaze___ Slow Glaze___
Custom "vary fire" program_____

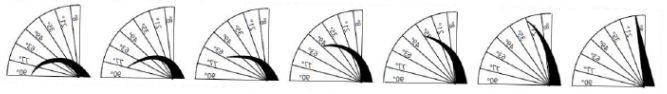
Target Cone #_____ Preheat Time: _____ Soak/ Hold Time: _____

Time	Temperature	Comments (e.g. notified Marc Drader re: _____)
Firing Start	°	
	°	
	°	
	°	
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	°	
Firing Finish # hours_____/ display	°	

Potters must be present in the Workshop for the first and last two hours of firing

Finished Cone Pack Positions
(Circle the images that best match your cone packs and identify the cone # [5,6,7 or 07,06,05]) OR DRAW:

Upper Peeps 

Lower Peeps 

Please document which shelves you used (numbers are on the underside and edge of each shelf) and any cleaning performed.

Shelf Number	Cleaning Required and Actions

☐ Removed bottom shelves to vacuum Kiln. Bottom posts are in correct position.
For safety reasons, please restack the shelves in their slots rather than leaving them out by the kilns. Thanks!

Apr 1, 2022 v10

WPW Manual Electric Kiln # 1 Firing Log

Date_____ **Potters Firing**_____

Load size: Heavy___ **Medium**___ **Program: Bisque** ___ **Glaze**___

Do not fire a partially filled kiln as it is hard on the elements and wastes electricity.

Check list:

- ☐ **Correct target cone (Cone 06 Bisque ☐ or Cone 6 Glaze ☐) is in kiln sitter**
- ☐ **Timer is set to at least 1 hour longer than expected firing time**
- ☐ **Cone packs are in correct position for viewing through peepholes**
- ☐ **Kiln venting and room exhaust fans are turned on**
- ☐ **Peepholes closed once any preheat is finished**

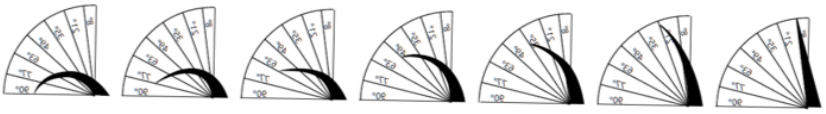
Time	**Setting** (LO, 2, 4, 6, HI)	**Temperature** (Pyrometer gives an approximate reading)	**Comments** (e.g. notified Marc Drader re: _____)
		°	
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New January 2023: Potters must be present in the Workshop for the majority of time of manual firing. You may leave for a brief period of time once the temperature is set to HI but must return for the last 2 hours of firing. When you are running it as a preheat for greenware on the LO setting. You may leave and return when the preheat time is finished. An example of this is as follows: In the evening, set timer to maximum time, then turn on bottom switch ONLY to low. Open the peeps and turn on the fans. This process may also be done over several hours before firing. Once you start to adjust the program and increase the settings, you must stay in the studio.

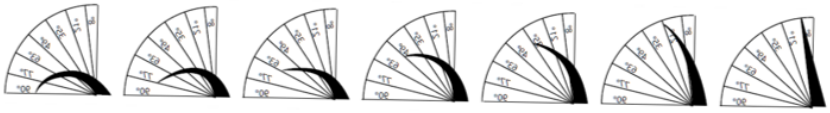
*** See page 2 for documenting Cone Pack results and use/maintenance of kiln shelves.

Finished Cone Pack Positions
(Circle the images that best match your cone packs and identify the cone # [5,6,7 or 07,06,05])

Upper Peeps



Lower Peeps



Please document which shelves you used (numbers are on the underside and edge of each shelf) and any cleaning performed.

Shelf Number	Cleaning Required and Actions

☐ Removed bottom shelves to vacuum Kiln. Bottom posts are in correct position.
For safety reasons, please restack the shelves in their slots rather than leaving them out by the kilns. Thanks!

9. What to do/who to notify if there are any kiln maintenance issues

For any electronic/ electric issues such as kiln fan switch not turning on, kilns underfiring or failing, elements coming out of the channels in the kiln sides, thermocouples being damaged or bent you must notify the Electric Kiln and Studio Maintenance person and copy the Electric Kiln Convener so that any kiln bookings affected by this can be cancelled. Their contact information is posted on the kiln room bulletin board at the bottom of the stairs.

For any kiln furniture (shelves, posts, stilts) issues such as breakage, glaze left on shelves or problems in the room contact the Electric Kiln Convener.

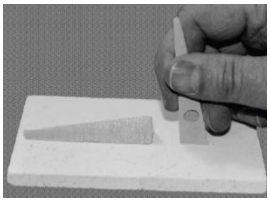
10. Making Cone Packs

We use the regular large Orton cones to make cone packs for bisque and glaze. Three cones are used: the Guide cone lets you know that the ware is reaching maturity, the Firing Cone tells you when you've reached the optimum point, and the Guard Cone is the back-up. The small kiln-sitter cones are only used for the manual kiln (Kiln 1). When cones soften and bend, this indicates that a certain amount of heat has been absorbed by the cone and pots being fired.

Properly used, pyrometric cones can be used to:

- Indicate when a firing is completed, when placed inside a kiln where they can be observed through a peephole.
- Verify the firing uniformity of a kiln, side to side and top to bottom.
- Check the temperature measured by an electronic kiln controller.
- Make a permanent record of the firing (on our WPW Kiln firing log).

Making new cone packs is the responsibility of each user.



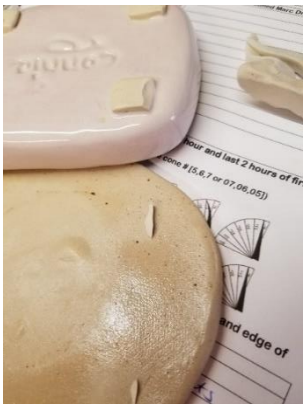
- The regular large cones that we use need to stand at a very specific angle in the kiln
- The "front face" of the cone should be 8° backwards from vertical as shown in this Orton photo
- This is the exact angle that the cone sits naturally when placed on a flat surface, also known as the bending face, indicating the direction in which the cone will bend when that temperature is reached
- This is the proper mounting angle for the cone, so when they are inserted into the clay plug, you need to maintain the same angle. This means that the flat edge with the cone number will be at a slight angle, visible from the front but appearing to lean slightly towards the left side of the clay plug. This allows the cones to fall diagonally and independently, not on top of the cone that melted prior to this one. The flat edge should be at the back
- We make our cone packs to fall to the left, so the order of cones is from coolest (L) to hottest (R).
- When finishing the cone pack, use a pin tool to poke holes into the clay plug all around the cones to allow even drying and to prevent the clay from cracking.





Supplies for Cone Packs:
Clay is available in a plastic container under the stairs in the kiln shelf maintenance area in the electric kiln room.
Orton cones are upstairs in the glaze room, first upper cupboard to the right of the outside door. Instructions are posted on how to separate the cones to prevent breakage. If you finish a box, please notify the Glaze Room Convener by email to order more.

11. Kiln Cookies



Kiln cookies are clay slabs that are placed under your glazed pottery pieces to protect your kiln self from drips of glaze, but also protect your pottery from any glaze left on the kiln shelf by previous users. Our Cone 6 cookies should only be used for up to Cone 6 firing.
Kiln cookies can also be used under large flat pieces of pottery as they fire. Big flat items like platters, plates, and tiles have a tendency to bend or warp when they are fired. This is because there is a lot of internal tension in large flat surface areas as they fire. Sometimes pieces will break under the tension.
Placing a cookie under a plate or a similar item helps with heat distribution during firing. The cookie acts as a buffer between the plate and the kiln shelf. If anything breaks or warps, it is likely to be the kiln cookie rather than your pottery. (Gabriel Kline Amazing Glaze 2018; https://thepotterywheel.com/what-are-kiln-cookies/)
If the glaze drips, the cookie must be large enough to catch the glaze but not too big that it takes up too much room in the kiln.
If glaze from your pot melts onto the cookie, your pottery will fuse to the cookie. If you're lucky, you may be able to chip the cookie off your pottery. Sometimes, it doesn't appear that there has been a glaze run, but the cookie is attached. Don't force it apart; take it home with the cookie attached and it often separates within 24-48 hours as it cools.

- Cookies usually last 4-5 firings before they bloat (bubbles form in the clay)
- A sheen of clear glaze as shown below can develop during firing without actual drips occurring, and as per the photo, can destroy your work
- Please discard kiln cookies that are bloated, have glaze drips or the sheen of clear glaze. DO not reshelve.

Making Kiln Cookies
We depend on volunteers to make cookies for the kiln room. Please make some every few firings while you are waiting for the kilns.

- Cookie cutters and cone 6 clay are found with the other tools under the stairs in the kiln room
- Make a variety of sizes in round and square shapes
- Small cookies should be one-eighth of an inch thick (use all of the boards in the slab roller
- Larger ones need to be thicker, removing one or two boards in the slab roller
- Use a pin tool to cut a smooth edge if needed to avoid sharp points
- To dry, they must be kept flat to prevent warping, important because the cookie needs to be stable enough to support your pottery. Place a drying board approximately the same size over the first board on which you have placed the cookies. Place 2 or 3 more boards on top to weigh it down, and label "WPW Cookies."
- Before placing on the "please fire me shelf" or firing yourself, wipe the edges smooth with a damp sponge.

12. Cleaning of the Kilns, Shelves and Posts

The following equipment for shelf and post maintenance is found in the work area under the stairs in the kiln room:

- ✓ Instructions
- ✓ bright light
- ✓ pink all-purpose grinding stone
- ✓ Dremel with diamond tip
- ✓ diamond sanding bat for posts
- ✓ hammer, chisels, protective goggles
- ✓ kiln wash & brushes
- ✓ wheeled cart to place shelf on

Vacuuming the kilns after each use is mandatory and is described in the kiln firing instructions. A sign also describes safe use of the central vac above the outlet just outside the furnace room.
After either type of firing: If you find shelves with loose or cracking kiln wash please clean these up using the flat side of the pink all-purpose grinding stone as per instructions that follow.

a. Cleaning after bisque firing



We encourage people to flip the kiln shelves during a bisque firing to prevent warping or sagging of the shelves (especially kiln 2). Because there is no glaze involved in bisque firing, whichever surface you have used to place the greenware on should be clean when you are unloading. However, kiln wash may have adhered to the underside (without kiln wash) of the shelf from the posts and should be scraped clean and dusted to prevent it falling into your glazed pots and ruining them.

b. Cleaning after glaze firing

It is imperative that each user clean up any drips or spatters of glaze that has been deposited on the kiln shelves. The following tools can be used:



- **Chisel and Hammer –** Use gloves and eye protection for drips of glaze – (can also be used outdoors)
  - Change the angle of the chisel – 45 degrees or steeper
  - Use the corner of the blade more than the flat part.
  - Hammer the chisel lightly to get an edge and then whack it: tap, tap, whack

  Sharpen the chisels as needed – Use gloves and eye protection
  - Turn the grinder on in the slab roller room
  - Hold the chisel against the grinder wheel so that the flat side is facing you
  - Keeping it flat work back and forth so that it sharpens evenly





- **Dremel tool** – Can be used indoors. Use gloves, mask, eye protection and water bottle sprayer
  - To prevent the tip of the dremel tool getting hot quickly and to reduce dust - spray down the shelf with a water bottle first, dry off your hands.

- Press and hold the lock button to loosen the collar and put in a bit. Try different bits to see what works for you.
- Hold the tool like a pen. Brace your elbow and use your second hand if necessary. Be careful not to cover the vents in the tool.
- Use *medium or high speed*. Keep the tool moving fairly quickly. Keep spraying area with water to keep the tip cool but keep your hands dry.
- Create an edge around a glaze spill and then chip at the edges with a chisel.
- Keep brushing the dust away so you can see how you're doing.

- **All-purpose grinding stone**
  - Use **flat bottom** for cleaning crackled kiln wash off shelves and abrading spatters of glaze
  - DO NOT USE CORNERS of stone to dig out SPOTS—this will cause deep grooves in the kiln wash
  - These must be removed to prevent them from becoming airborne and ruining glazes in other firings. Use dry then wash shelf thoroughly before applying kiln wash.



c. Kiln Wash

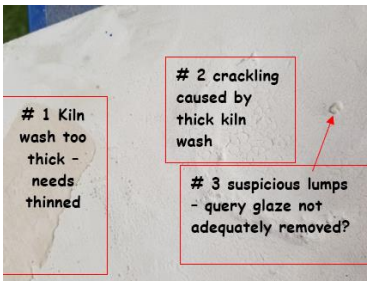
Making kiln wash is the responsibility of the Kiln Convener. The John Britt recipe used since 2020 is as follows:

- Alumina Hydrate - 50%
- Calcined EPK 25%
- Kaolin (EPK) - 25%
- G-200 Mahovir Potash Feldspar 1.5%
- Bentonite 5% (added to prevent clumping in bottom of pail)

Thin to the consistency of milk, use on dampened kiln shelves in a thin coat repeated 2 or 3 times if needed.

- If it is too thick (#1) it will just crackle when fired (#2).
- Crackled kiln wash must be cleaned off and any suspicious lumps (#3).

Damp sponge the shelf first to remove any dust. Apply 1 thin coat with a brush or roller and let dry for 24 hours before applying a second coat (if needed). Leave a note with the shelf asking the next person using the kilns to apply a second coat. They can store it away once that is dried.



d. Annual cleaning

Updated July 2023:

Process: Once per year, to thoroughly clean the kiln shelves and posts, and the kiln room, the Electric Kiln Convenor, prior to the call for Spring Cleaning which goes out in March, will provide the Spring-Cleaning Convenor with:

1. A list of all members who currently fire or are apprenticing with the Electric Kilns so that the Spring Cleaner Convenor can choose from when assigning jobs.
2. Indicate any specialty that those members have, for example shelf sanding and grinding, kiln maintenance, post grinding etc.
3. The number of 3 hour shifts/ Days/times based on the days set aside for Spring Cleaning. The first part of the cleaning week might be best due to the dust created.
4. The Electric Kiln Convenor will provide the Spring-Cleaning Convenor with a list of the jobs that are required and the total number of people needed.
5. The Electric Kiln Convenor does not need to supervise on a daily basis unless they choose to.

The Spring-Cleaning Convenor will then assign members to these roles and shifts with specific duties assigned. Any issues that may arise should be dealt with through the Spring Cleaner Convenor.

Cleaning: Outdoor Shelf Maintenance may include stripping kiln wash off any shelves that require complete stripping, removing glaze spills left by users, reapplying kiln wash, sanding all posts to remove adhered kiln wash.

The following tools and methods are used:



i. **Stripping shelves of kiln wash and removing glaze spills that cannot be removed with chisel and hammer:**
   - New July 2023: Belt sander with belt for masonry (shown with belt being loaded at right). This tool has been re-introduced as it is less damaging to the shelves than the angle grinder, but effectively removes old kiln wash. The belts become dull quite quickly. Use your own respirator mask, eye and ear protection (3M ear protection available in the supplies under the stairs).
   - Angle grinder for large areas of glaze spills with a sanding disc for masonry, not wood– use outdoors only. Use your own respirator mask, eye and ear protection (3M ear protection available in the cupboard). This will take off most glaze spills



     - The right hand needs to hold the body of the machine and the thumb works the switch. Place the left hand on the side handle. This is the safe way to proceed.
     - Use this tool on its edge
     - Keep it moving – it digs deep if it is still
     - Grinding happens fairly smoothly with the wheel engaged at around 45 degrees or less (Denise U. suggests 30°). This presents more surface area to the work surface, making for a smoother job

ii. **Kiln Post Maintenance**
   - Diamond bat on wheel for cleaning kiln posts – it should f it snuggly over the pins. Use gloves, mask, and eye protection. **Note-** Discard posts which are chipped at the corners or cracked so they do not stand up straight.



     - Always use with water to maximize tool life & control dust
     - Use medium fast speed. Sand bits off both ends of posts. Set the post down to check if it sits flat or still has protrusions to grind off
     - Be careful not to change the height of any posts
     - Carefully clean splash pans to remove all sharp debris.

13. "Please Fire Me" shelves NEW content Jan 2023

Potters who are not able to fire independently can leave small numbers of pots on the "Please Fire Me" shelves in the kiln room. Potters will fit in any pieces that they can as they fire their kilns. There is no charge for firing test tiles, but potters may ask for a monetary contribution for other pieces, based on the suggested pricing grids below.

A communication tool has been created to leave information for any pots left on the "Please Fire Me" shelves, either for "Please fire Me" or for community kilns and can be found either on or near the "Please Fire Me" shelves. The forms contain instructions on when to leave the pots etc. and a reminder that pots not picked up after 3 months from the blue shelves outside the damp room will be donated to Celebrate Summer.

When leaving pots, remember that members do treat other people's work with care, but breakages do happen.

The suggested pricing grid is for both Community Kilns and "Please Fire Me" so that pricing is consistent, fair and transparent.

The final amounts for firing are based on:

- # of pieces and how they fit on the shelf
- # of shelves required for total kiln
- Height of posts required for your pots.

Firing Communication Request

Place a small slip of paper with your name inside each pot

☐ **"Please Fire Me"** - for small numbers of pots: members will fire in their kilns as they are able

☐ **Commmunity Kiln** - book by emailing communitykilns@waterloopotters.ca saying how many pieces, how tall and how wide they are. **If not booked, may not get fired.** Only leave Community Kiln pots on "Please Fire Me" shelf one week before booked firing date or on numbered kiln shelf 24 hours before firing.

☐ **△06 Bisque** ☐ **△6 Glaze**

Date: _____

Your name: _____

Cell: _____

Email:

Number of pots:

Any special instructions?

If you want cookies or stilts, set them up under the pieces on a small bat.

After firing:

Fired by: _____

Date left on Blue shelves: _____

**** Pots not picked up within 3 months will be donated to Celebrate Summer**

"Please fire me" prices are at the discretion of the person firing, and payable to that person who will contact you with a price and instructions for payment.

Bisque Firing

Post Height Shelf	2-3" or less	4-5"	6-8"	9" or taller
1/4	$1.25	$2.08	$3.18	To be determined
1/2	$2.50	$4.17	$6.25	
3/4	$3.75	$6.25	$9.43	
Full	$5.00	$8.34	To be determined	

Glaze Firing

Post Height Shelf	2-3" or less	4-5"	6-8"	9" or taller
1/4	$1.70	$2.83	$4.25	To be determined
1/2	$3.40	$5.67	$8.50	
3/4	$5.10	$8.50	$12.75	
Full	$6.80	$11.34	To be determined	

14. Community Kiln Program NEW content Jan 2023

This is a service provided to fire small numbers of pots for members who have not yet completed their apprenticeship for firing who need to have pots completed on a schedule rather than using the "Please Fire Me" shelf method. Suggested pricing is as per the grid in the "Please Fire Me" section above, pro-rated at the discretion of the coordinators using those guidelines. An invoice payable to WPW will be sent to you following the firings.

The kilns are booked ahead for the next calendar year, so that people who need to use the Community Kiln firings can plan ahead and book with the Community Kiln Conveners. The schedule is posted in the Electric Kiln room, but the dates can be subject to change. To be sure, always check the Kiln Calendar on the website which will be kept up to date. The email address to simplify communications about Community Kilns: communitykilns@waterloopotters.ca.

Community Kiln firings can be used for Apprenticeship experience for those members who are looking to complete their Apprenticeship.

15. Apprenticeship Program

The WPW kiln firing course constitutes the mandatory orientation component and leads to the apprenticeship program of mentored firings. A hands-on workshop to clean and maintain kiln shelves and furniture will also be required. The class component could also be open to members who have already completed their apprenticeship but would like to learn more.

Part 1: Electric Kiln Course

Mandatory Pre-learning: watching the video re: applying glazes to make sure it does not run.

Course Hours:

- Friday: 6-9 p.m.
- Saturday: 9 a.m to 4 p.m.
- Sunday 1-4 p.m.

Course Content: Theory and Practicum of firing both Bisque and Glaze kilns; safety and responsibilities including a post-test.

Cost is $130 plus tax
Cancellation fee of $20.00

Minimum 5 students, Maximum 8 students
Please register through Member Classes on the website.

This course is the mandatory educational component for the newly standardized Apprenticeship Program for Firing the Electric Kilns. class is also open to members who have already completed their apprenticeship but would like to learn more.

Please bring glazed bisque ware and green ware to the class.

(Mandatory for all members wanting to fire Electric Kilns not already approved)

Part 1: Electric Kiln Course

Course Hours: Saturday 11 hours and Sunday 2 hours – includes loading plus break of 2-3 hours before finishing firing on Saturday

Cost: Per WPW standard member costing –

- Fees: recommended cost of $99 per student, with WPW subsidy if less than 5, and minimum 3 required to run
- Costs:
  - Instructor payment for 13 hours Saturday and Sunday @ $30 = $390
  - Photocopies of handouts, about 10 pp each, ~ $3 per student
  - Firing 1 glaze and 1 bisque kiln = $34 + $25 = $59
  - Glaze for one kiln = $25
  - Total for 5 students = $489

Time need to change	Activity	Instructor/Notes
Before Sat	• Students & Instructors bring glazed and bisque ware to studio for the Saturday class.	Instructor book 4 kilns
Saturday 0800	• Load a bisque kiln – Start kiln	Observe – demo
0800	• Load a glaze kiln -Start kiln	Observe - demo
	• Electric kilns safety and room venting. • Discussion- why venting is important	Observation Already in kiln room
	• Introduction to the electric kiln & safety • How to use the WPW Electric Kiln manual • Ensure coils are clean and not bulging • Discussion of manual kiln	Students observe and start and stop the kilns using the controllers.
	• How to prepare the green ware for bisque firing.	Students use own green ware to prep
	• How to make cone packs • Cone pack placement.	Make cone packs Bsq & Glaze
	• How to load green ware • Where to place ware - how close to coils, handles placement, thermocouple	Demonstration above – finer points – is it wet?
	• Unloading Bisque - inspection	What to look for Unload Kiln
	• Inspection & Cleaning of kiln after bisque firing including shelves.	Clean actual shelves with equipment - outside
	• Importance of record keeping, how to record information	
	• Kiln firing temperature & kiln atmosphere environment. • Kiln firing schedules for bisque and glaze. • When/when not to use sand	
	• Discussion on washing & waxing the pots in preparation for glaze firing. • Do's & Do Nots of glazing	Students use own bisque ware
	• How to load a glaze kiln	finer points – e.g is the glaze wet, double checking bottoms and glaze line.

	• Use of cookies for glaze ware	
	• Unloading Glaze Inspection	What to look for Unload Kiln
	• Inspection & Cleaning of kiln after glaze firing including shelves (how to)	Clean actual shelves with equipment – outside hands on
	• Payment for firing & glazing	
Sunday 1700	• Unload both kilns, discussion of ware and cleaning	Students participate in unloading and cleaning
	• Payment for firing & glazing	Check student work

Part 2: Apprenticeship

- Members who have been firing independently for more than three years and fire frequently may serve as mentors. An updated list of members who may fire can be found in the front of the apprenticeship binder.
- **Part 3: WPW Electric Kiln Manual**

Part 4: Process for Certification

- Completion of the Electric Kiln Course
- Successful participation in test/quiz at end of course
- Successfully complete four electric kiln firings including one manual kiln firing with approved mentors within the workshop. Once three firings are completed (one bisque, one glaze in computer controlled kilns and one manual firing) over a minimum of two firing days, the apprentice firer must be in charge of the final firing—including booking, loading, firing, completing paperwork, cooling, unloading and subsequent cleanup (such as repairing shelf damage).
- Recommendation from the Mentor
- Recommendation from the Electric kiln Convenor
- Approval sign-off by the Executive.

While not mandatory, it is beneficial for newly approved Electric Kiln users to fire with a buddy for several firings, as we recognize that this apprenticeship cannot cover all possible firing challenges.

Part 5: In addition, a formal re-approval process of members who have not fired in over a one-year period will be required.

Responsibilities of electric kiln convenor re: Apprenticeship

- Register the members who wish to complete an apprenticeship.
- Start a record to record their firings and teacher
- Receive the apprentice's paperwork, which is a record of completed firings and checklist of firing procedure. This record sheet is in the binder on the shelf in the kiln room.
- Check with the teaching members about any concerns before the convenor's approval that the member has completed the required number of firings, and passes the completed paperwork to the Executive as notification
- Run a kiln furniture maintenance workshop for apprentices if timing does not coincide with annual cleaning, or cleaning cannot be done at time of WPW course due to weather.
- The paper record of approval is then kept on file by the electric kiln convenor at the workshop.

Responsibilities of teaching member

- Members are not obliged to mentor others, but they are encouraged to do so in the spirit of the
- cooperative.
- A member who agrees to teach apprentices may work with up to a maximum of 3 apprenticing members during one firing day
- Initial each item on the apprentice's record of firing and firing checklist.
- Ensure they address each of the items on the checklist before, during and after firing.
- Report any safety or comprehension concerns to the electric kiln convenor.
- When supervising the final firing, ensure that the apprentice safely and competently fires the kiln according to the policies of the workshop.

Responsibilities of Apprentice

- Notify the electric kiln convenor of their intention to complete the apprenticeship program by participating in the required course, mentored kiln firings and kiln furniture maintenance workshop
- Approach experienced members to teach/mentor them. Identify mentors who are firing by using the electric kiln booking calendar on the website
- Maintain their *Record of Firing for Electric Kiln Apprentice* which is kept in the binder on the shelf in the kiln room
- Inform the electric kiln convenor that they have completed their apprenticeship.

Record of firing for electric kiln apprentice

Revised July 2023

Name of Apprentice:_____

Date of WPW Kiln Course:_____WPW Course Teacher's signature:_____

Computer Kiln Firings:

Item	Bisque Firing	Glaze Firing	Final Demonstration Firing
Firing Date			
Teaching Member			
Checked bottom of kiln for debris or silica sand left by previous member; vacuum if necessary			
Assisted with loading the kiln: selected height of posts, carefully handled and positioned ware, discussed when to use kiln cookies for glaze kiln, avoided damage to kiln walls and thermocouple(s)			
Checked shelves for glaze or cracked kiln wash on top, adhered kiln wash on underside, cleaned and brushed before using, inspected for cracks, reporting damaged shelves to Kiln Convener			
Added pieces from Please Fire Me shelves if possible			
Turned on kiln fan and room fans; ensured outer door can be opened; removed flammables near the kiln			
Programmed computer controller			
Checked that latch is closed on lid and peeps are closed			
Started and kept the kiln log			
Present for first hour and last two hours of firing			
Prepared Cone Packs and or kiln cookies with Teaching Member			
Discussed problem solving if something goes wrong: reviewed error codes in manufacturer's manual or website; demonstrated how to check breaker in Electric Panel			

Discussed cooling the kiln; when safe to open peeps and crack lid; when to unload			
Inspected shelves and posts for damage; removed glaze drips and oxide spatters, loose kiln wash; applied kiln wash; re-stacked shelves for storage			
Vacuumed kiln removing bottom shelves; vacuumed around kiln floor.			
Ensured that all pots are removed from kiln room; pots from Please Fire Me shelf go onto Blue Shelves in cubby room with completed paperwork			
Discussed booking kiln online			
Discussed making payment for kiln			

Manual Kiln Firing

Item	Completed
Firing Date	
Teaching Member	
Checked bottom of kiln for debris or silica sand left by previous member; vacuumed if necessary	
Added correct cone to kiln sitter before loading	
Assisted with loading the kiln: selected height of posts, carefully handled and positioned ware, discussed when to use kiln cookies for glaze kiln, avoided damage to kiln walls and thermocouple(s)	
Checked shelves for glaze or cracked kiln wash on top, adhered kiln wash on underside, cleaned and brushed before using, inspected for cracks, knows to report damaged shelves to Kiln Convener	
Added pieces from Please Fire Me shelves if possible	
Turned on kiln fan and room fans; ensured outer door can be opened; removed flammables near the kiln	
Closed latch on lid and peeps	
Set timer to at least 2 hours beyond expected time frame and turned time sitter switch to LO to start kiln	
Started and kept kiln log, checking the pyrometer for temperature	
Each hour turned the elements up by 2, so LO, 2, 4, 6, HI.	
Present until setting is on high and returned for last two hours of firing	
Prepared Cone Packs and or kiln cookies with Teaching Member	
Discussed problem solving if something goes wrong: reviewed error codes in manufacturer's manual or website; and how to check breaker in Electric Panel	

Discussed cooling the kiln; when safe to open peeps and crack lid; when to unload	
Inspected shelves and posts for damage; removed glaze drips and oxide spatters; applying kiln wash; re-stacked shelves for storage	
Vacuumed kiln removing bottom shelves; vacuum around kiln floor.	
Ensured that all pots are removed from kiln room; pots from Please Fire Me shelf go onto Blue Shelves in cubby room with completed paperwork	
Reviewed booking kiln online	
Reviewed making payment for kiln	

Participated in a hands-on workshop to clean and maintain kiln shelves and furniture
Signature of Teacher:_____ Date:_____

Approved as a firer:

Electric Kiln Convener's signature:_____ Date:_____

Executive Approval:

President's signature:_____ Date:_____

This document is to remain in the apprenticeship binder during the apprenticeship and upon completion.